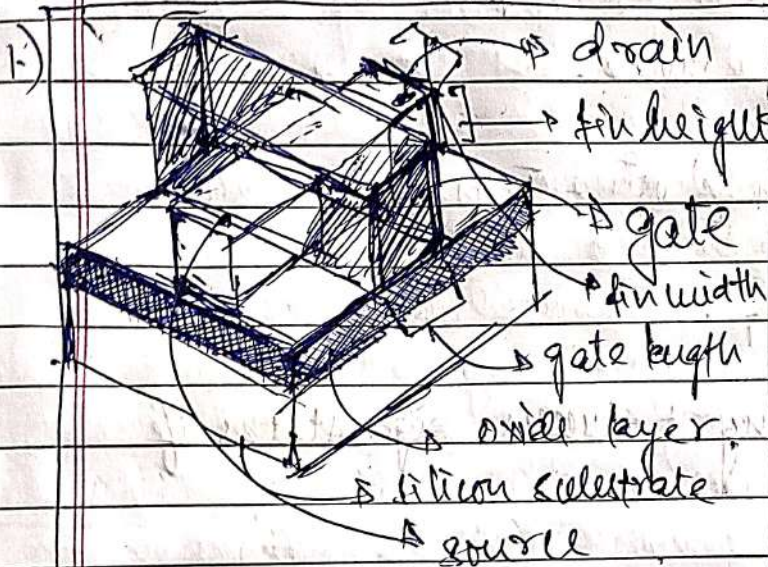


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Nanoelectronics [H.W. - 2]



SOI - FinFET.

SOI FinFET is a 3D multi-gate transistor built on a silicon-on-insulator substrate. Its structure consists of a thin, vertical silicon "fin" [the channel] standing on buried oxide layer, which isolates it from the bulk silicon handle wafer. The gate electrode wraps around

3 sides of this fin (tri gate architecture) contacting the channel on top & both sidewalls.

The current conduction mechanism is different.

⊗ In planar, current flows horizontally via channel below a flat gate interface leading to significant SCEs. In FinFET, channel flows vertically along the fin, gate exerts electrostatic control from 3 sides. This suppresses E coupling b/w source & drain which is the root cause of SCEs.

Reason for IMPROVEMENT ON I_{on} & I_{off}

① Improved I_{off} → wrap around gate → better control, preventing from being influenced by the gate voltages. This leads to steeper subthreshold swing meaning transistor can switch off more abruptly. This significantly ↓ subthreshold current.

② Improved I_{on} : Effective width: $(2 \times H + W)$. This allows larger drive current per unit. Furthermore, good control allows for the use of lighter channel doping, which reduces impurity scattering, enhancing mobility $\uparrow \mu_n$

→ I_{off} (off state leakage) ↓

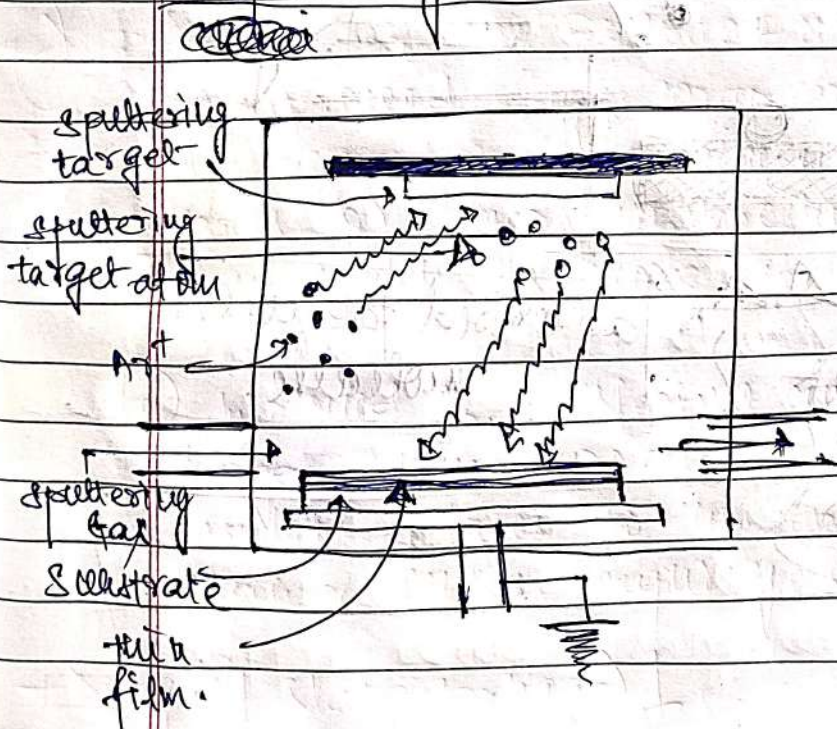
→ I_{on} (on state current) ↑

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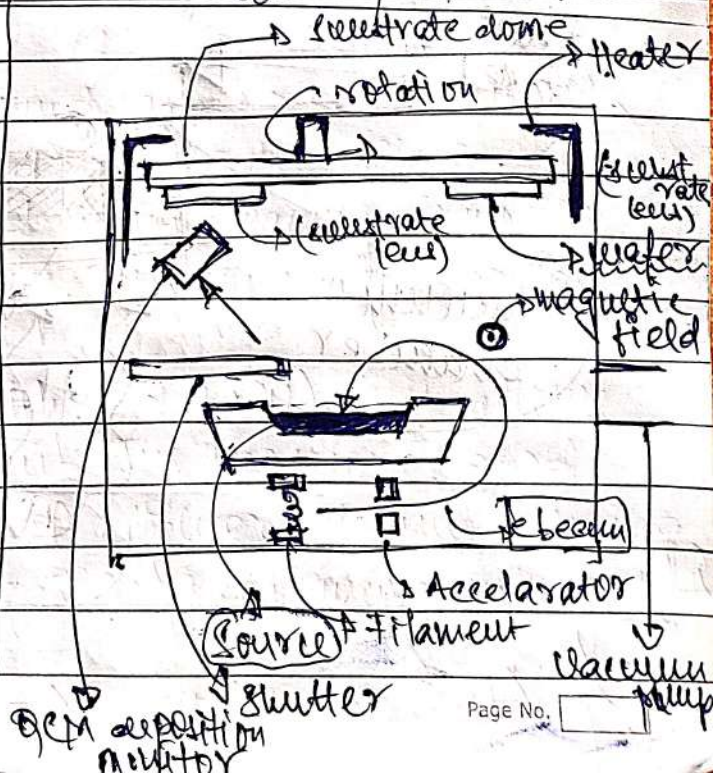
(A) Physical Vapour Deposition.

- (1) PVD utilizes solid / liquid source materials that are physically vaporized through processes like sputtering or e-beam evaporation.
- (2) The vaporized atoms or molecules travel ballistically in a vacuum chamber and condense on the substrate.
- (3) Reaction control is primarily physical & directional, governed by parameters such as power, pressure & substrate orientation.
- (4) The deposition rate R_{pvd} can be modelled as proportional to the vaporization flux, but the process exhibits poor step coverage due to its line-of-sight nature, resulting in non-conformal films with thickness variations over topography.
- (5) Film growth is characterized by columnar microstructures and may require adhesion layers for improved bonding.

Sputtering PVD



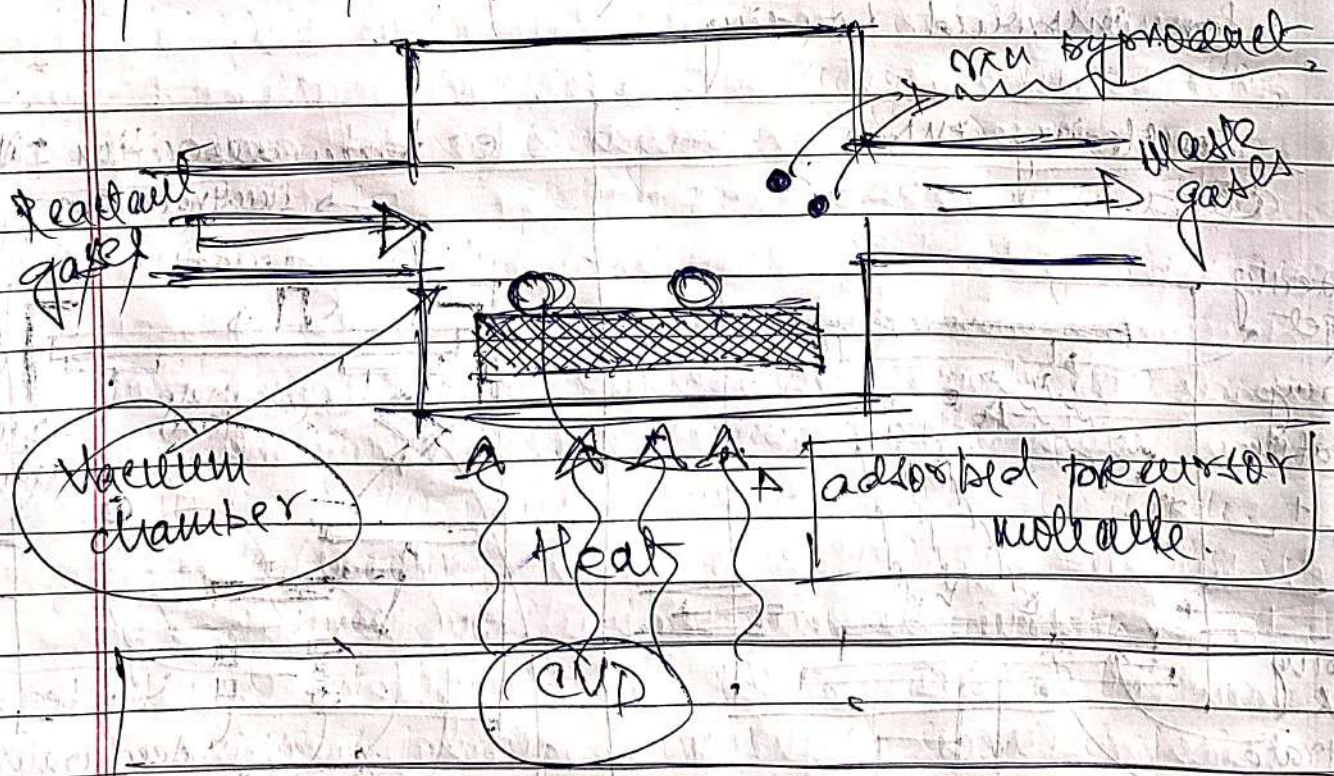
e-beam deposition PVD.



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chemical vapor deposition

- ① CVD employs gaseous precursors (eg: SiH_4 , WF_6) that undergo chemical reactions on a heated substrate surface.
- ② The process relies on surface-mediated chemical reactions, where precursor molecules adsorb, react and form a solid film, with volatile byproducts desorbing & being removed by gas flow.
- ③ The deposition rate $R = k_s \cdot c_s$ ($k_s \rightarrow$ surface coefft, $c_s \rightarrow$ precursor concentration)
- ④ CVD enables conformal film growth due to surface diffusion & reaction, allowing uniform coverage over complex geometries.
- ⑤ However, it requires high temperatures to activate chem rxns, and process is influenced by mass transport & rxn kinetics, which leads to UNIFORMITY CHALLENGES or particle formation.



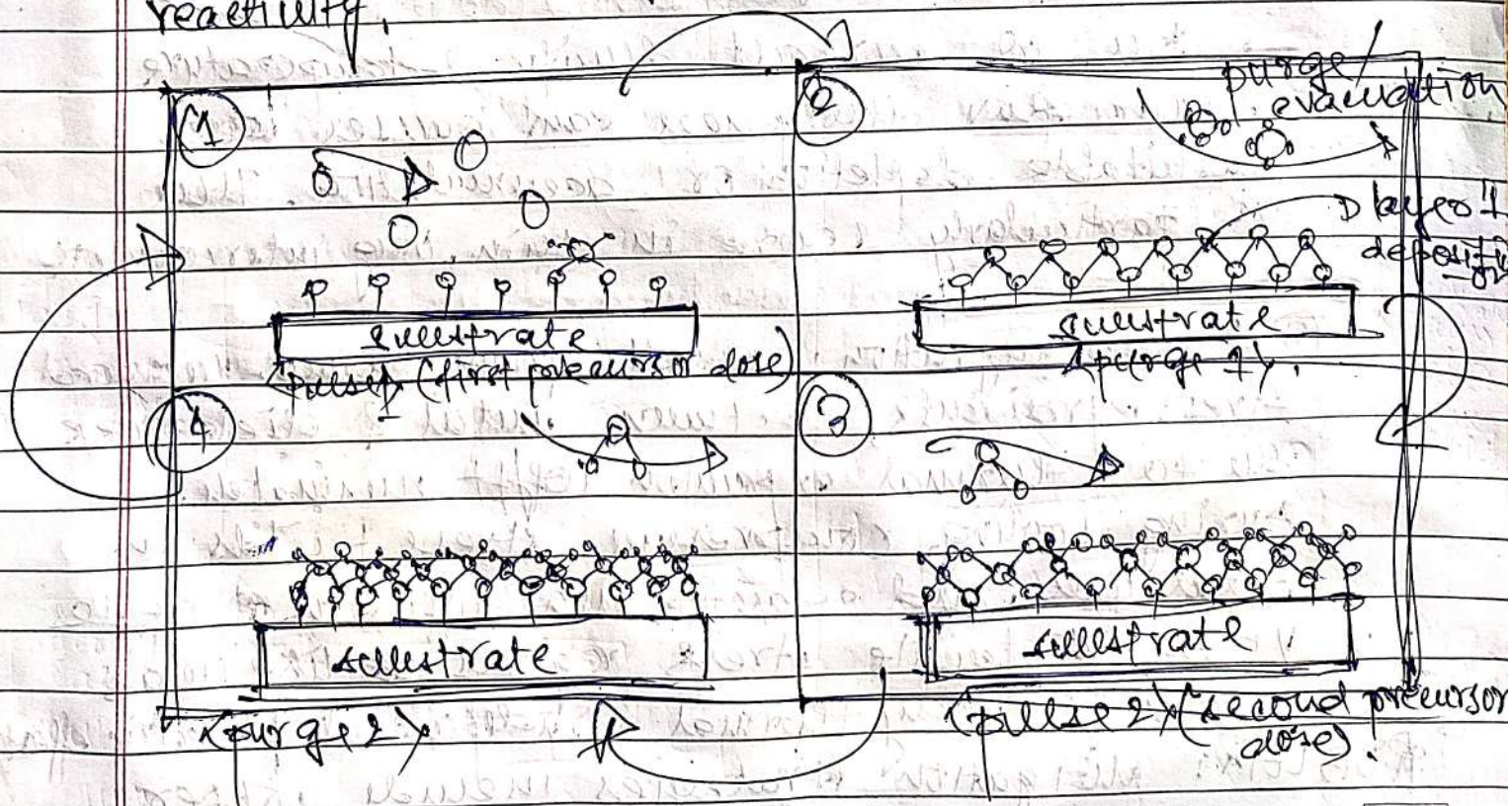
(4)

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Atomic Layer Deposition (ALD)

- ① ALD also uses gaseous precursors but operates via sequential, self-limiting surface rxns.
- ② Each precursor is introduced separately, with purging steps in between, ensuring that only one monolayer forms per cycle.
- ③ This results in precise thickness control governed by number of cycles & growth per cycle.
- ④ The self-limiting mechanism provides exceptional conformality & uniformity, even in high aspect ratio structures.
- ⑤ ALD operates under milder/moderate temps and offers superior control over film composition and interface quality.
- ⑥ However its deposition rate is significantly lower than PVD & CVD and requires compatible precursor chemistry with sufficient volatility & reactivity.



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2) Interconnect scaling significantly impacts reliability through 3 mechanisms:

① RC delay: As interconnect dimensions shrink, the resistance increases due to reduced cross-sectional area ($R = \rho \frac{L}{A}$), while capacitance also increases due to tighter spacing ($C = \epsilon A_d$) leading to higher RC delay that dominates circuit performance.

PROBLEM: slows signal performance. Increases power consumption CV^2f and causes crosstalk.

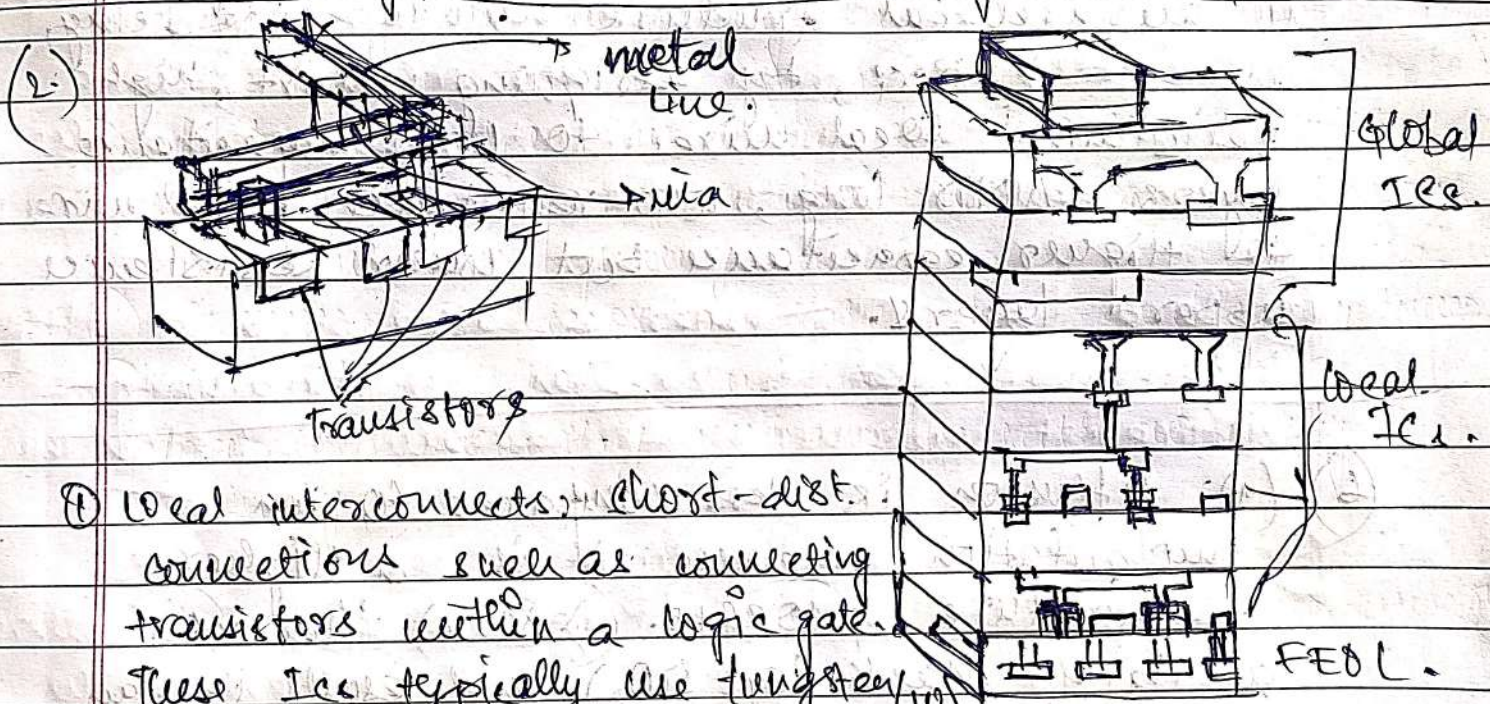
② Electromigration: occurs when high current density causes metal atoms to drift in the direction of e^- flow. This creates vacancies (voids) that can open circuits or hillocks (deposits) that can short adjacent lines.

PROBLEM: due to current density & temperature atomic flux divergences can cause ~~void~~ inevitable depletion or accumulation. This is particularly severe in thin, long interconnects.

③ Stress migration: results from thermo-mechanical stress gradients, between metal & dielectrics due to thermal expansion coeff mismatch. During cooling, compressive stress builds in metal lines, and relaxation via diffusion creates voids at tensile stress regions like vias & corners. Purely thermal & independent of electric eff.

PROBLEM: Mitigation strategies include copper

interconnects with barrier layers (Ta/TaN) to reduce electromigration, implementing capping layers (SiN, CoWP) to control diffusion, and employing low- κ dielectrics to minimize RC delay and thermal mismatch. Layout optimization like redundant vias and shorter lines help distribute stress & current. Thermal management through design and annealing processes reduces both electromigration & stress migration effects.



- ① Local interconnects: short-dist. connections such as connecting transistors within a logic gate. These ICs typically use tungsten (W) due to high melting point. Their dimensions scale with tech node.
- ② Semiglobal ICs: larger block connections. larger area, to reduce resistance over moderate distances. Copper (Cu) is the primary material here.
- ③ Global ICs: long distance routing. This includes distributing power (V_{DD}) & ground, and connecting major blocks. Copper is also the primary material here, just thicker & wider to minimize resistive losses.

(4) Vias: vertical connections made up of tungsten (W) plugs, that electrically connect all the above metal layers, barrier layers ($Ta-TaN$) are used with copper to prevent diffusion into the surrounding dielectrics.

(5) FEOL: The IC stack is built on the substrate AFTER the Front-End-Of-Line (FEOL) transistor fabrication.

The hierarchical structure minimizes delay, as $\tau = \sum R_i C_i$, by assigning short, high resistance local wires to low capacitance layers AND long, low resistance global wires to higher capacitance but lower resistance upper layers.

(6) (a): Diffusion is less compared to ion implantation because it is a higher temperature process ($> 800^\circ C$) that uses a hard mask (e.g. SiO_2) and results in an isotropic (lateral) dopant profile. It cannot ~~independently~~ independently control the dopant concentration and junction depth, unlike ion implantation.

(b)

(1) Parameter	Diffusion	Ion implantation
① Dopant control precision	Low, depends on gas concentration & time.	Highly precision dose control.
② Junction depth	Deep (μm scale) controlled by time & temp.	Shallow (can be $< 50\text{nm}$) controlled by ion energy.
③ Process temperature	High ($800 - 1200^\circ\text{C}$)	Low (room temp for implant)
④ Crystal damage	Damage free.	Creates crystal defects. Requires annealing.
⑤ Equipment complexity	Simple (furnace-based)	Complex (ion sources, mass analyzers, beam optics)

(2) ~~Optical~~ Projection lithography is with a step - 4 - repeat system.

Steps to obtain a final pattern on a silicon wafer using optical photolithography:

① Surface preparation: The wafer is cleaned to remove contaminants. An adhesion promoter like HMDS (Hexamethyldisilazane) is applied to ensure the photoresist sticks properly to the substrate.

② Photoresist coating: A positive tone photoresist is spin coated onto the wafer surface, forming a uniform, thin light-sensitive polymer layer.

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- ③ Soft Bake: The wafer is heated at a lower temperature (eg, 90°C) to evaporate the solvent from the resist, solidifying the film.
- ④ Alignment & exposure: The wafer is aligned under a stepper. UV light is projected through a photomask onto the resist. In positive resist, the exposed regions become soluble.
- ⑤ Development: The wafer is immersed in a developer solution (like TMAH: tetramethylammonium hydroxide). This removes the exposed, soluble areas of the positive resist, transferring the mask pattern onto the resist layer.
- ⑥ Hard Bake: The wafer is baked again at high temperature to harden the remaining photoresist, improving its durability & etch resistance.
- ⑦ Etching: A dry etching process, such as Reactive Ion Etching (RIE), is used. The patterned photoresist acts as a mask, protecting the underlying material. The RIE process anisotropically removes the unmasked material, creating vertical sidewalls, and transferring the pattern into the substrate.
- ⑧ Photoresist removal: The remaining photoresist mask is stripped away using a plasma or a chemical process, leaving the final patterned structure on the silicon wafer.

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(a) Polysilicon ~~materials~~ interacts poorly with high- κ dielectrics due to:

1. Fermi level pinning: High κ dielectrics (eg HfO_2) have fixed charges that pin the polysilicon Fermi level, altering the effective work function and increasing threshold voltage. (V_{th})
2. Phonon scattering: Polysilicon/high- κ interface ~~degrades~~ degrades carrier mobility via remote Coulomb scattering.
3. Depletion effects: Polysilicon depletes at the interface, reducing gate capacitance & overdrive voltage.
- (b) In the gate last process, the high- κ dielectric & metal gate are deposited after source-drain annealing, avoiding thermal degradation. In gate-first (poly-Si, SiO_2) the gate stack experiences high temperature steps, causing V_{th} shifts due to dopant diffusion and interface reactions. Gate-last enables independent metal work-function tuning for NMOS (eg TiAlN) & PMOS (eg TiN), optimizing V_{th} without thermal budget constraints.